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THE ORDER OF QUESTIONS IN A SURVEY ON CITIZEN SATISFACTION WITH PUBLIC SERVICES: LESSONS FROM A SPLIT-BALLOT EXPERIMENT

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Surveys of citizen satisfaction with local public services have become widespread, with the results increasingly used to reorganize services, to allocate budgets, and to hold managers accountable. But evidence from a split-ballot experiment that we conducted suggests that the order of questions in a citizen survey has important effects on reported satisfaction with specific public services as well as overall citizen satisfaction. Moreover, the correlations of specific service ratings with overall satisfaction, and thus the identification of key drivers of overall satisfaction, also turn out to be highly sensitive to question order. These findings are in line with research on priming and question order effects in the survey methodology literature, but these effects have not been carefully examined before in the context of citizen surveys and public administration research. Policy and management implications of these finding are discussed.

INTRODUCTION

Most citizens have an opinion about their local public services, even when they are infrequent users (Van de Walle 2009). We also know from the literature that citizens' overall evaluations of public services tend to be quite different from their evaluation of specific services (Grunow 1981; Klages 1981; Goodsell 1983; Kelly and Swindell 2003), and that the link with objective quality is uncertain (Van Ryzin 2004, 2008a). A possible explanation for this divergence is the salience of the object of evaluation. Specific local public services have a high salience for most citizens, because they use these services, know people who use them, or they in other ways encounter these services on an almost daily or weekly basis. Evaluating public services in general is harder because the object of measurement is ambiguous. Yet, minor changes in levels of citizen satisfaction with public services are often taken at face value by policy-makers and can become the basis for important decisions about service budgets and service delivery.

When a respondent answers a question such as 'How satisfied are you with service X', the answer may depend on a very wide range of elements: recent experience, media stories, information from peers, experience in the past and so on. If, to use one of the questions used in the analysis later in this article, you were to ask the authors of this article how satisfied we are with the overall functioning of public services in our town, our answers will depend on our recent experience, past experience, hearsay, or a recent crisis or event. We would probably compare our experiences with public services to experiences with other types of services. As public administration scholars, we would probably add some of our professional knowledge to the formulation of our answers. And if you would insist, we would probably tell you that we find answering such a question quite hard. We are satisfied with some public services, less so with others. There are many public services we have never used ourselves. And there are likely to be services we have not even heard of. We may even answer the question in a strategic way, and say we are not satisfied as a way of either increasing the budget spent on public services, or expressing our disapproval with the administration.

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Some questions may also be very ambiguous, yet most respondents will formulate an answer anyway. The concept 'public services' is quite general, and we cannot be certain that all respondents answering a question on public services will think about the same set of services. Some opinions may be influenced by services that are not public services at all. Other general opinions may be strongly influenced by the experience with a single public service. And in other cases stereotypical views of government may influence the answers (Van de Walle 2004a).

Minor changes in a questionnaire can have an important impact on how citizens formulate their satisfaction judgements about (public) services (Schul and Schiff 1993). Such question order effects have received considerable attention in the survey research literature (McFarland 1981). For example, priming occurs when prior items (primes) in a questionnaire affect subsequent items (targets) (Tourangeau *et al*. 2000). Priming studies attempt to alter the relative accessibility of material, facts, beliefs and so on for respondents when answering survey questions (Tourangeau *et al*. 2000; Gaines *et al*. 2007). Survey context and question order are included in most survey methods handbooks, yet the topic receives surprisingly little attention from researchers who do not in the first place write about survey methods. There are some exceptions though (DeMoranville and Bienstock 2003).

When analysing service satisfaction data, policy makers and public administration scholars often tend to take the data as a given and do not consider context effects. In a way, this is normal, because studying context effects is not their first objective. However, ignoring the existence of such effects may have important implications for the interpretation of satisfaction survey data. Levels of satisfaction can easily go up or down due to factors that have in fact little to do with the performance of the service. This is especially problematic where such satisfaction scores are used to measure performance, to decide about resources, or to evaluate individual authorities, managers, or street-level workers.

In this article, we use data from a split-ballot survey experiment to test whether and how the order of specific and general questions about local public services influences reported levels of satisfaction with these services. We also test if question order affects the correlations between specific service ratings and overall citizen satisfaction. We first review the evidence from recent research on citizen perceptions of public services, and some of the factors influencing these perceptions. We then introduce the reader to the methodological literature on priming in surveys, and the effect of question order in questionnaires. This section is subsequently used to reflect on the use and value of experiments in public administration research. We continue by introducing our dataset and the two versions of the questionnaire used in the split-ballot experiment. The body of the article presents the findings, and we conclude with a discussion of the implications of our findings for the study of citizen perceptions of public services, and for the practice of using citizen satisfaction indicators in policy making.

WHERE DO PERCEPTIONS OF PUBLIC SERVICES COME FROM?

Research into satisfaction with public services has a long history in urban research (Stipak 1977, 1979, 1980), and we have recently seen a renewed interest in the study of citizen satisfaction with local public services (Van Ryzin *et al*. 2004; Swindell and Kelly 2005; James 2009), and with public services in general (Bouckaert *et al*. 2005; Van de Walle *et al*. 2008). Elements of this research have focused on the value of satisfaction surveys for

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assessing the quality of public services (Miller and Miller 1991), and the link between objective and subjective measures of performance (Brown and Coulter 1983; Parks 1984; Swindell and Kelly 2000; Kelly and Swindell 2002; Van Ryzin 2008a; Van Ryzin *et al*. 2008).

It has been assumed that evaluating the quality of a local public service is relatively straightforward for citizens. After all, local services have a high salience for most citizens. They use these services, know people who use them, or they in other ways encounter these services on an almost daily basis. The evaluations of different public services could therefore be expected to be highly diverse, in line with different levels of service quality. Overall evaluations of local services would then be based on these specific evaluations. However, research has consistently found that overall evaluations often tend to be quite different from specific service evaluations. The most common finding has been that asking about specific services generally results in higher overall satisfaction, whereas more general evaluations of government services tend to be more negative (Grunow 1981; Klages 1981; Goodsell 1983; Kelly and Swindell 2003). In addition, it appears that citizens without recent experience with public services, or even without any experience at all, have opinions about specific public services.

One observation in service satisfaction research is that citizens' opinions are influenced by factors other than service quality (Katz *et al*. 1977; Goodsell 1983; Van de Walle 2004b; Van Ryzin 2004; Van Ryzin and Immerwahr 2004; Van Ryzin *et al*. 2004; Bouckaert *et al*. 2005; Orr and West 2007). In satisfaction surveys, general opinions about government, politics, or bureaucracy often appear to influence evaluations of individual public services. More general opinions on government have an impact on opinions about specific institutions (Van de Walle *et al*. 2005). It appears that 'an underlying general predisposition in evaluating specific local services appears to exist. The service evaluation domains all share a major common component' (Stipak 1977, p. 51). There is a common component in citizens' attitudes towards public services and institutions, or a predisposition (Kampen *et al*. 2006). Certain abstractions or belief systems introduce certain intercorrelations between attitudes towards various local services (Stipak 1977). Therefore, citizens' evaluations may sometimes be influenced less by concrete experience than by abstract orientations towards government, or 'symbolic attitudes', or predispositions (Sears 2001; Orr and West 2007).

These effects are likely to be stronger when the object of evaluation is of a more general or ambiguous nature, and weaker when the object of evaluation is quite specific. In other words, when citizens are asked to evaluate government services in general, their opinion is more likely to be influenced by this 'common component' or 'predisposition'. If the object of evaluation is very specific, the impact of this predisposition will be weaker. The respondent is then more likely to accurately recall and judge objective service quality. For example, Van Ryzin *et al*. (2008a) found a strong correlation between New York City's street cleanliness scorecard and citizens perceptions of street cleanliness, with general attitudes towards government having only a limited influence.

In an experiment in two small cities in Florida, Benton and Daly (1991) manipulated the question order in a survey of municipal services. One question was a general evaluation of municipal services; the other questions were evaluations of specific services. They summarize their findings as follows: 'When the service-specific questions were asked first, 71.9 per cent of the respondents evaluated the overall quality of services as 'very good' or 'good,' while only 63.8 per cent did so if the general question preceded the service-specific questions.' (Benton and Daly 1991, p. 641). Communication Canada, then a Canadian government agency, found in its surveys *Communication Canada* that respondents' ratings of government's overall performance depended heavily on the

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location of this question in the questionnaire. When preceded by questions assessing specific government performance in various policy domains, ratings of overall government performance where considerably higher than when this question appeared near the beginning of the questionnaire (Fasiolo 2002). In research on trust in government, Lock *et al*. (1999) found higher trust among respondents who were first asked about specific government operations. Respondent's inclination for giving consistent answers makes it hard for them to rate government negatively if they have just approved of a number of specific government operations. Reminding respondents of certain government tasks and outputs influences their response behaviour. In a survey of 1,248 inhabitants of Flanders (Belgium) in 2002, Van de Walle *et al*. (2005) found that overall satisfaction with public services was a factor more strongly related to preceding general questions on trust in government, than to subsequent questions on satisfaction with specific public services.

These effects appear to have a differential effect on different groups of respondents. A higher level of education, better information, exposure to political information and so on contribute to belief system differentiation. This means that for respondents who are better informed and higher educated, the common factor or predisposition has a weaker impact on specific opinions (Stipak 1977; Benton and Daly 1991; Van de Walle 2004b). As a result, these respondents differentiate better between different public services. This shows that citizens do not necessarily share a common point of reference when evaluating public services (Brudney and England 1982, p. 128).

THEORETICAL BACKGROUND: PRIMING EFFECTS

These examples demonstrate that simple changes in a questionnaire may have an important impact on how citizens will evaluate public services. Question order effects in survey research have received considerable attention in the literature (McFarland 1981). The findings reported in the previous section are good examples of what survey methodologists call priming effects. Prior items (primes) in a questionnaire affect subsequent items (targets) (Tourangeau *et al*. 2000, p. 176). The presence of certain attitude objects triggers an evaluation (Sears 2001). The literature generally distinguishes between assimilation (carryover) and contrast (backfire) effects. Assimilation occurs when respondents are asked a general question, and their answer is influenced by a preceding specific question; information from the prime is included in the considerations for the answer in the target. Contrast occurs when the information contained in the prime is excluded from consideration in the target – this tends to happen especially when a specific item follows a general one (Sudman *et al*. 1996). The literature further identifies two broad kinds of effects. One effect is on the speed of responses, whereby priming facilitates access to answers to subsequent questions on a similar topic. The other is on the direction and strength of the response, where the prime has an effect on the content of the answer. In this article, we will exclusively focus on the latter.

When people are asked to answer a question in a survey, they often do not have an answer at hand. Surveys are not mechanical instruments. Answers respondents give depend on a series of cognitive processes (Sudman *et al*. 1996). Answering a factual question is generally relatively simple, but even such purely descriptive information asked in surveys may be inaccurate (Wentland and Smith 1993, pp. 19–42). The problem naturally becomes even more important when attitudes rather than facts are measured. The theory on priming 'is based on the assumption that gathering information requires time and effort, and therefore the citizen will use readily available information' (Yeric and

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Tood 1996, p. 131). Primes provide such readily available information, and they guide the respondent to generalizing to the subsequent item, to contrast, or to open a new box of considerations. The primes affect the accessibility of answer to later questions (see also Zaller 1996; Tourangeau *et al*. 2000, p. 177). When in a survey situation, respondents want to reduce uncertainties (Tversky and Kahneman 1982), and therefore they don't do an exhaustive search for an answer.

Priming is more likely to occur when the question is relatively hard to answer, or when it may relate to many different issues. When respondents do not hold a highly structured belief system on a certain issue, they construct an opinion based on more general beliefs, values or predispositions (Zaller 1996). Attitudes on an issue may be hard to retrieve (Tourangeau *et al*. 2000), or may simply not exist. Giving answers that are not accurate is therefore not necessarily intentional (Wentland and Smith 1993), and contradictions may be quite normal. Respondents use a wide range of available information to answer survey questions, including feelings, emotions beliefs, general values, prior judgements about an issue and so on (Tourangeau *et al*. 2000, p. 179). The result of this is that responses to attitude questions are unstable (Tourangeau *et al*. 2000, p. 181).

When measuring attitudes towards public services, the effect of priming is likely to be strong when we measure overall attitudes towards public services, largely because the object of measurement is so ambiguous. General and ambiguous questions following more specific ones are likely to be influenced by the preceding ones, because certain issues have been made more salient. Without the prime, the information for answering a question may be inaccessible for respondents, or the respondent may have difficulties selecting the relevant elements for answering the question.

The precise effect of priming is disputed. Priming effects are not always found, and the effects of a prime are sometimes limited to the item immediately following the prime and not beyond (Sudman *et al*. 1996). The salience of the issue is also important, and priming effects are likely to be lower for questions on high salience issues, because the information is more easily available. A question about a dramatic failure of specific public service is likely to influence the subsequent question on public services in general, but less likely to influence a subsequent question on a quite different specific public service. In the former case, there is assimilation, because the target is inclusive: 'public services' as a category also includes the specific failing public service. In the latter case, there is contrast because the target is narrowly defined (Sudman *et al*. 1996, p. 119).

Primes do not have a uniform influence on affects, but are also mediated by characteristics of the respondent. These include the respondent's awareness of the topic and the structure of the respondents belief system about the topic (Zaller 1996). Yet, the evidence on whether more knowledgeable respondents are less susceptible to context effects is mixed (Sudman *et al*. 1996, p. 103).

EXPERIMENTS IN PUBLIC ADMINISTRATION

Survey methodology research has devoted considerable attention to priming. Priming studies attempt to alter the relative accessibility of material, facts, beliefs and so on for respondents when answering survey questions (Tourangeau *et al*. 2000, p. 175). Such survey experiments have become quite popular in social research because they can rigorously demonstrate cause and effect, and because they can easily be embedded into other pieces of research. But they are of course not without problems, and the results can sometime be difficult to interpret because of the complexity of questionnaires and the subtlety of context effects (Gaines *et al*. 2007).

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The use of experiments in public administration is limited, especially when compared with disciplines such as psychology, where they are quite common, and political science, where experiments are rapidly gaining popularity (Gerber and Green 2000; Druckman *et al*. 2006; Kam *et al*. 2007; Morton and Williams 2010). In public administration, some publications using experiments can be found, but the numbers remain very small. Bozeman and Scott (1992) give an overview of the use of experimental methods in the public administration discipline. Examples of articles based on experimental designs include studies of administrative discretion (Scott 1997), decision-making (Weiss 1982; Bretschneider *et al*. 1988; Coursey 1992; Landsbergen *et al*. 1992; Landsbergen *et al*. 1997; Knott *et al*. 2003; Nutt 2006), ethical decision-making (Wittmer 1992), or performance assessment (Moynihan 2006). Experimental designs increase the internal validity of the study and its demonstration of cause-effect relationships. In addition, experiments come with important practical (cost-saving) advantages (Kam *et al*. 2007). Experiments also come with problems of external validity, but this is less the case in most split ballot experiments because they tend to be based on fairly large samples of a general population.

Bozeman and Scott (1992) give a number of reasons for the relative absence of experimental methods in public administration research, including its focus on groups or organizations as the unit of analysis, rather than individuals, and the unwillingness of public administration scholars to trade in external validity to gain stronger internal validity. In addition, many public administration research topics do not easily lend themselves to an experimental approach because of ethical reasons or practical constraints. Still, the use of experiments in the discipline, both laboratory experiments and field experiments, is likely to increase along with developments in neighbouring disciplines

A SPLIT-BALLOT EXPERIMENT ON SATISFACTION WITH PUBLIC SERVICES: DATA AND SET-UP

Split-ballot is an experimental technique where groups of respondents to a survey receive different versions of the questionnaire on a random assignment basis. The order of questions or other elements of the questionnaire constitute the experimental treatment that is manipulated (Sniderman and Grob 1996). The technique is often used to test and evaluate measurement instruments, and to test the effect of the use of neutral categories, don't know options, the order of response lists, question wordings and so on (Clarke *et al*. 1999; Linde and Ekman 2003). For this article, we conducted a split-ballot experiment in which we manipulate the order of questions in a citizen satisfaction survey to test for priming and other context effects, both on the level of reported satisfaction and on covariation or correlation among variables.

The data for our analysis come from an online survey conducted in October 2004 of participants in the CivicPanel project (formerly eTownPanel project), an Internet research panel of approximately 3,592 participants (at the time of the study). Online panels are increasingly being used for research in social, health and marketing research (Wansink and Sudman 2002). The CivicPanel project is a university-affiliated panel created to provide a general population of US volunteers to participate in online surveys about public and civic affairs sponsored by nonprofit organizations and academic researchers. Volunteers are recruited from various online postings and notices, including Google advertising, Craigslist.org, and the open directory Dmoz.org, and are not a random sample of the population. Nevertheless, validity studies using the panel suggest that weighted results from the panel approximate traditional telephone polls based on probability sampling of

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the US population (Van Ryzin 2008b). However, in our analysis of experimental results, we have chosen to use unweighted data in order to ensure efficiency and precision on our calculation of test statistics for purposes of causal hypothesis testing. Although weighting data by demographic and other variables is appropriate for descriptive analysis of complex survey data (Lee and Forthofer 2006), it is not standard practice for causal hypothesis testing in the context of experimental data (Brown and Melamed 1990).

Invitations were sent via email to the entire panel, and a total of 1,638 completed the online questionnaire, for a panel response rate of 45.6 per cent. Random numbers were used to assign respondent to two versions of the questionnaire, Version A and Version B. A total of 830 respondents completed Version A of the questionnaire, 808 respondents completed Version B.

Figure 1 shows how the order of questions in the two questionnaires varied. Both versions begin with a series of general questions that ask the respondent to rate his or her city or town as a place to live, followed by a question regarding confidence in local political leaders. Version A then asks respondents to rate their overall satisfaction with local government services (7-point scale), followed by ratings of specific government services, such a parks, police, schools, transportation, and so on (5-point scale). Version B, in contrast, first asks respondent to rate specific services, then follows this series with the overall satisfaction questions. These questionnaire versions are fairly typical of the kind of options chosen by public administrators and policy analysts when implementing a citizen survey.

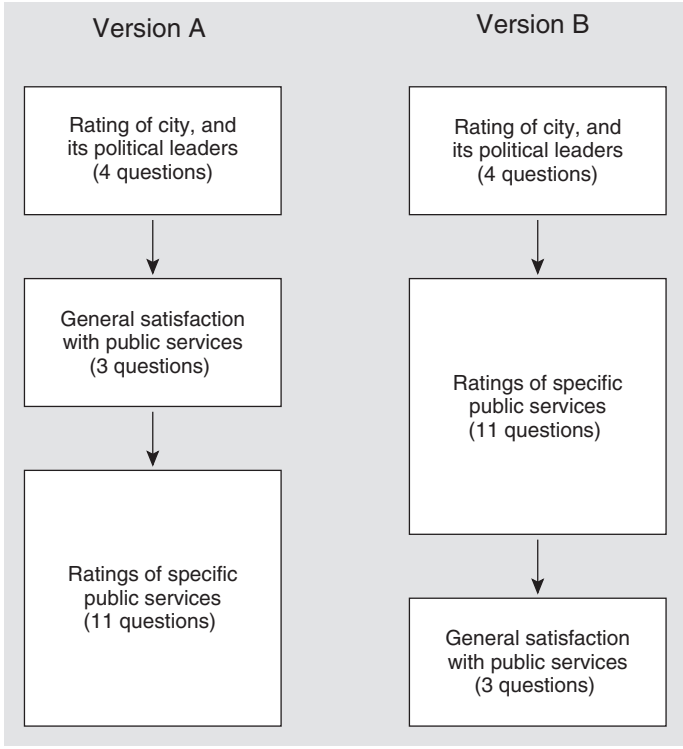


FIGURE 1 *Design of the split-ballot experiment*

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TABLE 1 *Demographic comparison of the samples*

	Version A	Version B	Chi-sqr	DF	p-value
N	830	808			
Male	27.9	27.1	0.12	1	0.73
Female	72.1	72.9			
White	85.6	87.3	5.11	3	0.16
Black	6.2	4.8			
Hispanic	2.3	3.6			
Asian	5.9	4.3			
Under 25	6.8	7.0	0.85	3	0.84
25 to 44	54.9	53.3			
45 to 59	32.7	33.1			
60 or older	5.6	6.6			
Less than HS	2.8	2.7	0.70	2	0.71
HS graduate	59.0	61.1			
College graduate	38.2	36.1			
Less than 25.000	16.5	17.7	1.38	3	0.71
25000 to under 50.000	35.6	33.7			
50000 to under 75.000	25.8	27.6			
75.000 or more	22.1	21.0			
Large city	25.3	27.2	0.90	3	0.83
Suburb	29.7	28.1			
Small city	32.2	31.6			
Rural area	12.8	13.1			

As table 1 shows, the two groups are comparable in terms of gender, race, age, education, income, and type of community, with no statistically significant differences on any of these variables. This confirms the validity of the random assignment and the statistical equivalence of the two experimental groups. It should be noted that the relatively high proportions of females and whites are characteristics of this particular panel (Van Ryzin 2008b), although a similar demographic pattern seems to characterize other voluntary online panels in the US (Berrens *et al*. 2003).

SUMMARY OF EXPERIMENTAL FINDINGS

We first compare the two groups in terms of their general satisfaction. The question asked respondents: 'Overall, how satisfied are you with the services provided by the local government where you live?' Responses were recorded on a scale from 1 'completely dissatisfied' to 7 'completely satisfied'. As figure 2 shows, satisfaction scores on this question differ significantly between the two groups. When this general question comes before questions about specific services (Version A), the score is significantly higher than when it follows questions about specific services (Version B). In other words, asking specific questions first seem to deflate overall satisfaction. These findings contradict earlier research that suggested that being primed about specific aspects of government performance (through the specific satisfaction questions) would increase citizen evaluations (see above; see also Lock *et al*. 1999).

When we look in more detail at the differences between groups across other variables in the survey, as shown in table 2, a more complex pattern becomes apparent. A first observation is that there are no significant differences between the answers to the opening questions in Versions A and B, which is reassuring and again confirms the comparability

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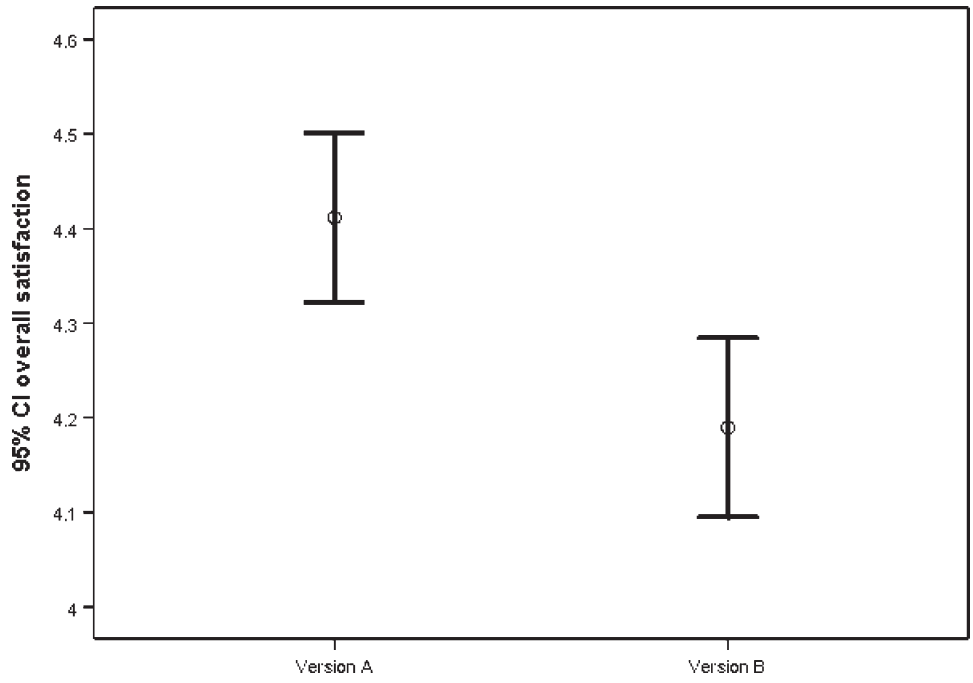


FIGURE 2 *Overall satisfaction in two versions of a questionnaire (1–7 scale)*

of the two samples. But it should be pointed out that these opening questions ask about the respondent's community and community attachment, perhaps priming some measure of pride in their community. In Version A, the overall satisfaction question perhaps reflects some of this positive priming.

Differences are also evident in the list of specific satisfaction questions. Satisfaction ratings of police protection, police-community relations, and fire protection (all uniformed services, it is worth noting), along with the ease of car travel, are all significantly higher when preceded by the general satisfaction questions (Version A). This may suggest an assimilation effect is occurring in Version A, whereby respondents inflate their specific service ratings after having just answered an overall satisfaction question, which itself appears to have been influenced by priming from the opening community attachment questions. The fact the uniformed services appear most influenced can perhaps be taken as evidence of this community attachment, or community pride, priming effect.

In this second part of our analysis, we test for question order effects on the covariation or correlations among variables. More specifically, we look at the individual effects and combined explanatory power of specific service ratings as predictors of overall satisfaction, and how these differ across both experimental groups. Priming theory suggests that when specific satisfaction questions (prime) precede a general satisfaction question (target), these primes will have an impact on the target, because the primes partly determine which elements respondents take into account when answering a target question. This implies that the pattern of important predictors of overall satisfaction will differ, depending on whether they precede or follow the overall satisfaction question. It also implies that a model of overall satisfaction, with specific service ratings as independent variables, will fit better when the general satisfaction question is asked at the end (as in Version B).

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TABLE 2 *Two sample t-tests*

	Version A mean	Version B mean	Mean difference	t	df	p-value
Opening questions						
Rating of city as a place to live	2.28	2.28	0.00	−0.03	1523	0.974
City has become better/worse	2.67	2.69	−0.02	−0.38	1522	0.708
Willingness to move to another city	2.55	2.55	0.00	0.01	1521	0.993
Confidence in the people running local government	3.05	3.07	−0.02	−0.53	1520	0.598
General satisfaction (Version A asked here)						
Overall satisfaction with local government services	**4.41**	**4.19**	**0.22**	**3.34**	**1503**	**0.001**
Local government services better/worse than expected	4.16	4.06	0.10	1.55	1522	0.122
Comparison of local government services with ideal	3.94	3.93	0.01	0.12	1516	0.904
Specific service ratings						
Cleanliness of streets and sidewalks	3.49	3.39	0.10	1.83	1522	0.067
Street and road maintenance	3.01	2.94	0.07	1.20	1521	0.230
Parks and playgrounds	3.53	3.42	0.10	1.87	1514	0.062
Public libraries	3.70	3.62	0.08	1.49	1520	0.136
Garbage collection	3.65	3.59	0.06	1.13	1518	0.260
Police protection	**3.59**	**3.43**	**0.16**	**2.84**	**1517**	**0.005**
Police-community relations	**3.39**	**3.24**	**0.15**	**2.63**	**1521**	**0.009**
Fire protection	**3.96**	**3.80**	**0.16**	**3.42**	**1513**	**0.001**
Public education (K-12)	3.43	3.34	0.09	1.63	1511	0.103
Ease of car travel	**3.41**	**3.25**	**0.16**	**2.54**	**1516**	**0.011**
Ease of travel by public transportation	2.94	2.92	0.02	0.27	1517	0.788
General satisfaction (Version B asked here)						
Overall satisfaction with local government services						
Local government services better/worse than expected		See above				
Comparison of local government services with ideal						

Note: Overall satisfaction was measured using a 1–7 scale, specific satisfaction on a 1–5 scale. Bold indicates statistically significant differences at the $\alpha = .05$ level.

Table 3 presents the regression results for both versions of the questionnaire, with the 11 specific service ratings as the independent variables and general satisfaction as the dependent variable. The R square differs notably between the two experimental groups. In Version B, where the general satisfaction item follows the specific service questions, the model fit is much better, in line with the theory and with our expectations.

Not only the R squares are different, but so too the pattern of coefficients on the various independent variables. When the general satisfaction question precedes the specific questions (Version A), the top five independent variables are, in order of importance (as determined by the betas): public education, street cleanliness, police-community relations, street maintenance, and garbage collection. When general satisfaction follows specific satisfaction (Version B), a quite different list of drivers emerges: police protection, public transport, police-community relations, street maintenance, and parks and playgrounds. It is interesting to note that the top two drivers in the Version B model are not even significant, or just borderline significant, in the Version A model.

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TABLE 3 *Regressions*

	Version A						Version B					
Model summary	**R-sqr = 0.502**						**R-sqr = 0.656**					
Coefficients	**B**	**SE**	**Beta**	**Top 5 betas**	**T**	**p-value**	**B**	**SE**	**Beta**	**Top 5 betas**	**T**	**p-value**
Cleanliness of streets and sidewalks	3.10	0.80	0.161	2	3.88	0.000	2.36	0.68	0.116		3.47	0.001
Street and road maintenance	2.15	0.66	0.125	4	3.26	0.001	2.62	0.62	0.134	4	4.24	0.000
Parks and playgrounds	1.23	0.69	0.067		1.78	0.075	2.66	0.67	0.130	5	3.98	0.000
Public libraries	0.41	0.66	0.022		0.62	0.539	0.91	0.61	0.045		1.49	0.137
Garbage collection	1.64	0.67	0.084	5	2.46	0.014	0.12	0.63	0.006		0.19	0.852
Police protection	1.23	0.85	0.065		1.45	0.147	4.48	0.80	0.228	1	5.62	0.000
Police-community relations	2.61	0.81	0.142	3	3.21	0.001	2.75	0.77	0.144	3	3.59	0.000
Fire protection	0.61	0.77	0.028		0.79	0.431	−2.19	0.71	−0.096		−3.07	0.002
Public education (K-12)	3.47	0.61	0.196	1	5.70	0.000	2.41	0.55	0.123		4.36	0.000
Ease of car travel	0.62	0.52	0.038		1.18	0.238	1.63	0.47	0.095		3.47	0.001
Ease of travel by public transportation	0.87	0.49	0.053		1.76	0.079	3.34	0.42	0.193	2	7.92	0.000
(Constant)	−8.42	2.73			−3.08	0.002	−16.61	2.30			−7.21	0.000

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These results demonstrate that question order is not just a minor consideration, but can lead to quite different policy implications. For example, a key driver analysis using Version A of the questionnaire would suggest that public education and street cleanliness were the most important determinants of citizens' overall satisfaction with local government performance. Version B of the questionnaire, however, would suggest that police protection and public transportation are the key drivers of overall satisfaction. Yet both sets of results come from the same population of citizens experiencing the same variety of services; the only difference is in the order of asking the questions.

IMPLICATIONS FOR USING SURVEYS ON PUBLIC SERVICES AND CONCLUSION

In this article, we conducted a split-ballot experiment in which we manipulated question order in a survey on citizen satisfaction with local public services. We changed the order of specific and general questions on satisfaction with local services to analyse the effect on both reported levels of satisfaction scores and on covariation or correlation among variables. Through this manipulation, we found that important priming effects occur in such satisfaction surveys. Not only do reported satisfaction levels change as a result of question order, but also the covariation among the questions, as demonstrated in a comparison between two multiple regression models.

Contrary to many traditional experiments, our split ballot survey experiment was trialed among a large and diverse group of participants and not among a narrow group with specific characteristics (for example, public administration undergrad students). While representativeness was not our purpose, this sample set-up lends additional generalizability to the findings. Still, it must be pointed out again that this study is based on a nonprobability sample and results are not statistically projectable to the US population. Further research using probability sampling would be desirable, as well as studies that look into how respondent characteristics influence the strength and direction of priming effects. Respondents who are more knowledgeable about public services or have more frequent exposure to certain public services, or respondents who are simply more interested in public affairs may react differently to the priming. Secondly, a number of minor issues in the experimental design need to be corrected, including a further reduction of confounding factors, and a harmonization of the satisfaction scales for the specific and the general questions (now respectively a 5- and a 7-point scale) to allow for direct comparison of absolute values.

In the article, we showed that question location in the questionnaire has an impact on the reported satisfaction levels and on relations between variables and hence on explanatory models. Somewhat surprisingly, and in contrast to what was expected from the prior literature, asking about specific public services did not result in higher overall satisfaction with local government. Rather, our results showed the opposite: the specific service ratings, when asked first, resulted in lower overall satisfaction with local government. In part, we attribute this effect to a complex interaction with the initial question in the questionnaire, which referenced community attachments and may have produced their own order effect when directly preceding the overall satisfaction question. Still, these results suggest that more experimental work should be done before, with particular attention to question order, before accepting the assumption that citizens have a more positive outlook on specific services than on the performance of government in general.

In a policy context, our finding of order effects leads to two practical questions: (1) what is then the 'real' or 'true' satisfaction score?; and (2) should general or specific questions come first in public service satisfaction surveys? The answer is not an easy one to provide.

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There may be no such thing as a 'true' satisfaction score, and absolute frequencies or means are relatively useless, precisely because they are so unstable and subject to method effects. Survey research handbooks generally recommend having general questions precede specific ones. In this way, answers to the general question are unprimed: 'While we cannot know with certainty whether the ''true'' responses are those which precede or those which follow the specific questions, logic suggests that unprimed responses are probably closer to the respondents' real feelings' (McFarland 1981, p. 213). There is another side to this argument though. While unprimed answers are likely to be closer to 'real feelings', placing the general satisfaction question first also comes at a risk of creating such feelings where they don't exist. It is not entirely unlikely that average citizens have no established notion of their overall satisfaction with local public services. Answers obtained in this way may then be unprimed, but could at the same time be an artifact because they do not reflect a pre-existing opinion. Maybe people don't have a 'true' opinion about many specific public services, or about public services in general (Zaller 1996).

A second argument for putting the general question second is that the regression model we presented in table 3 fits better when the general question comes second. This may suggest that the general opinion is more evidence-based in this case, but on the other hand, it can also be an artifact because the answers have been primed by questions on a relatively random selection of specific public services. Furthermore, the survey methods literature is inconclusive about question order effects, and methodological research on service satisfaction surveys has found that question order effects are not consistent across different services (DeMoranville and Bienstock 2003, p. 228). Question order is not a mechanical thing – questions provide context. There will always be some context effect, because we are dealing with human judgement (Sudman *et al*. 1996).

Our findings have important policy implications, especially because public service satisfaction scores are increasingly being used to motivate service reforms, budget allocations, and management accountability. Satisfaction scores cannot be compared across surveys unless questionnaires, samples, non-response patterns and context are similar. This should be common knowledge, but it is often ignored in policy-making. Even more important is our finding that even relatively minor changes to a citizen satisfaction survey may alter the explanatory model behind the answers. In our example, this meant that changing question order lead to the identification of a different set of key drivers of satisfaction and thus, eventually, policy priorities.

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REFERENCES

Benton, J.E. and J.L. Daly 1991. 'A Question Order Effect in a Local Government Survey', *The Public Opinion Quarterly*, 55, 4, 640–2.

Berrens, R.P., A.K. Bohara, H. Jenkins-Smith *et al*. 2003. 'The Advent of Internet Surveys for Political Research: A Comparison of Telephone and Internet Samples', *Political Analysis*, 11, 1, 1–22.

Bouckaert, G., S. Van de Walle and J.K. Kampen. 2005. 'Potential for Comparative Public Opinion Research in Public Administration', *International Review of Administrative Sciences*, 71, 2, 229–40.

Bozeman, B. and P. Scott. 1992. 'Laboratory Experiments in Public Policy and Management', *Journal of Public Administration Research and Theory*, 2, 3, 293–313.

Bretschneider, S., J.J. Straussman and D. Mullins. 1988. 'Do Revenue Forecasts Influence Budget Setting? A Small Group Experiment', *Policy Sciences*, 21, 4, 305–25.

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Brown, K. and P.B. Coulter. 1983. 'Subjective and Objective Measures of Police Service Delivery', *Public Administration Review*, 43, 1, 50–8.

Brown, S.R. and L.E. Melamed. 1990. *Experimental Design and Analysis*. London: Sage.

Brudney, J.L. and R.E. England. 1982. 'Urban Policy Making and Subjective Service Evaluations: Are They Compatible?' *Public Administration Review*, 42, 2, 127–35.

Clarke, H.D., A. Kornberg, C. McIntyre *et al*. 1999. 'The Effect of Economic Priorities on the Measurement of value Change: New Experimental Evidence', *The American Political Science Review*, 93, 3, 637–47.

Coursey, D.H. 1992. 'Information Credibility and Choosing Policy Alternatives: An Experimental Test of Cognitive-response Theory', *Journal of Public Administration Research and Theory*, 2, 3, 315–31.

DeMoranville, C.W. and C.C. Bienstock. 2003. 'Question Order Effects in Measuring Service Quality', *International Journal of Research in Marketing*, 20, 3, 217–31.

Druckman, J.N., D.P. Green, J.H. Kuklinski and A. Lupia. 2006. 'The Growth and Development of Experimental Research in Political Science', *The American Political Science Review*, 100, 4, 627–35.

Fasiolo, R. 2002. Trust (and some interesting findings from our most recent survey), personal communication.

Gaines, B.J., J.H. Kuklinski and P.J. Quirk. 2007. 'The Logic of the survey Experiment Reexamined', *Political Analysis*, 15, 1, 1–20.

Gerber, A.S. and D.P. Green. 2000. 'The Effects of Canvassing, Telephone Calls, and Direct Mail on Voter Turnout: A Field Experiment', *American Political Science Review*, 94, 3, 653–63.

Goodsell, C.T. 1983. *The Case for Bureaucracy: A Public Administration Polemic*. Chatham, NJ: Chatham House.

Grunow, D. 1981. 'Client-centered Research in Europe', in C.T. Goodsell (ed.), *The Public Encounter: Where State and Citizen Meet*. Bloomington, IN: Indiana University Press, pp. 223–41.

James, O. 2009. 'Evaluating the expectations Disconfirmation and Expectations Anchoring Approaches to Citizen Satisfaction with Local Public Services', *Journal of Public Administration Research and Theory*, 19, 1, 107–23.

Kam, C.D., J.R. Wilking and E.J. Zechmeister. 2007. 'Beyond the ''Narrow Data Base'': Another Convenience Sample for Experimental Research', *Political Behavior*, 29, 4, 415–40.

Kampen, J.K., S. Van de Walle and G. Bouckaert. 2006. 'Assessing the Relation between Satisfaction with Public Service Delivery and Trust in government: The Impact of the Predisposition of Citizens toward Government on Evaluations of its Performance', *Public Performance and Management Review*, 29, 4, 387–404.

Katz, D., B.A. Gutek, R.L. Rahn and E. Barton. 1977. *Bureaucratic Encounters: A Pilot Study in the Evaluation of Government Services*. Ann Arbor, MI: Institute for Social Research.

Kelly, J.M. and D. Swindell. 2002. 'A Multi-indicator Approach to Municipal Service Evaluation: Correlating Performance Measurement and Citizen Satisfaction across Jurisdictions', *Public Administration Review*, 62, 5, 610–21.

Kelly, J.M. and D. Swindell. 2003. 'The Case for the Inexperienced User: Rethinking Filter Questions in Citizen Satisfaction Surveys', *The American Review of Public Administration*, 33, 1, 91–108.

Klages, H. 1981. 'Das Verhältnis zwischen Staat und Bürgern in der Bundesrepublik Deutschland', in M. Buse and H. Burschmann (eds), *Bürgernahe Verwaltung und der Verwaltungsausbildung*. Baden-Baden: Nomos Verlaggesellschaft, pp. 24–43.

Knott, J.H., G.J. Miller and J. Verkuilen. 2003. 'Adaptive Incrementalism and Complexity: Experiments with Two-person Cooperative Signaling Games', *Journal of Public Administration Research and Theory*, 13, 3, 341–66.

Landsbergen, D., B. Bozeman and S. Bretschneider. 1992. '''Internal Rationality'' and the Effects of Perceived Decision Difficulty: Results of a public Management Decisionmaking Experiment', *Journal of Public Administration Research and Theory*, 2, 3, 247–64.

Landsbergen, D., D.H. Coursey, S. Loveless and R.F. Shangraw Jr. 1997. 'Decision Quality, Confidence, and Commitment with expert Systems: An Experimental Study', *Journal of Public Administration Research and Theory*, 7, 1, 131–57.

Lee, E.S. and R.N. Forthofer. 2006. *Analyzing Complex Survey Data*. London, Sage.

Linde, J. and J. Ekman. 2003. 'Satisfaction with Democracy: A Note on a Frequently Used Indicator in Comparative Politics', *European Journal of Political Research*, 42, 3, 391–408.

Lock, S.T., R.Y. Shapiro and L.R. Jacobs. 1999. 'The Impact of Political Debate on Government Trust: Reminding the Public What the Federal Government Does', *Political Behavior*, 21, 3, 239–64.

McFarland, S.G. 1981. 'Effects of Question Order on Survey Responses', *The Public Opinion Quarterly*, 45, 2, 208–15.

Miller, T.I. and M.A. Miller. 1991. 'Standards of Excellence: US Residents' Evaluations of Local Government Services', *Public Administration Review*, 51, 6, 503–13.

Morton, R.B. and K.C. Williams. 2010. *Experimental Political Science and the Study of Causality: From Nature to the Lab*. Cambridge: Cambridge University Press.

Moynihan, D.P. 2006. 'What Do We Talk About When We Talk About Performance? Dialogue Theory and performance Budgeting', *Journal of Public Administration Research and Theory*, 16, 2, 151–68.

Nutt, P.C. 2006. 'Comparing Public and Private Decision-making Practices', *Journal of Public Administration Research and Theory*, 16, 2, 289–318.

Orr, M. and D.M. West. 2007. 'Citizen Evaluations of local Police: Personal Experience or Symbolic Attitudes?', *Administration and Society*, 38, 6, 649–68.

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Parks, R.B. 1984. 'Linking Objective and Subjective Measures of Performance', *Public Administration Review*, 44, 2, 118–27.
Schul, Y. and M. Schiff. 1993. 'Measuring Satisfaction with Organisations: Prediction from Information Accessibility', *Public Opinion Quarterly*, 57, 4, 536–51.
Scott, P.G. 1997. 'Assessing Determinants of Bureaucratic Discretion: An Experiment in Street-level Decision Making', *Journal of Public Administration Research and Theory*, 7, 1, 35–57.
Sears, D.O. 2001. 'The Role of Affect in Symbolic Politics', in J. H. Kuklinski (ed.), *Citizens and Politics: Perspectives from Social Psychology*. Cambridge: Cambridge University Press, pp. 14–40.
Sniderman, P.M. and D.B. Grob. 1996. 'Innovations in experimental Design in Attitude Surveys', *Annual Review of Sociology*, 22, 377–99.
Stipak, B. 1977. 'Attitudes and Belief Systems Concerning Urban Services', *Public Opinion Quarterly*, 41, 1, 41–56.
Stipak, B. 1979. 'Citizen Satisfaction with Urban Services: Potential Misuse as a Performance Indicator', *Public Administration Review*, 39, 1, 46–52.
Stipak, B. 1980. 'Local Governments' use of Citizen Surveys', *Public Administration Review*, 40, 5, 521–25.
Sudman, S., N.M. Bradburn and N. Schwartz. 1996. *Thinking about Answers: The Application of Cognitive Processes to Survey Methodology*. San Francisco, CA: Jossey-Bass.
Swindell, D. and J. Kelly. 2005. 'Performance Measurement versus City Service Satisfaction: Intra-city Variations in Quality', *Social Science Quality*, 86, 3, 704–23.
Swindell, D. and J.M. Kelly. 2000. 'Linking Citizen Satisfaction to Performance Measures: A Preliminary Evaluation', *Public Performance and Management Review*, 24, 1, 30–52.
Tourangeau, R., L.J. Rips and K. Rasinski. 2000. *The Psychology of Survey Response*. Cambridge: Cambridge University Press.
Tversky, A. and D. Kahneman. 1982. 'Judgment under Uncertainty: Heuristics and Biases', in D. Kahneman, P. Slovic and A. Tversky (eds), *Judgment under Uncertainty: Heuristics and Biases*. Cambridge: Cambridge University Press, pp. 3–20.
Van de Walle, S. 2004a. 'Context-specific Images of the Archetypical Bureaucrat: Persistence and Diffusion of the Bureaucracy Stereotype', *Public Voices*, 7, 1, 3–12.
Van de Walle, S. 2004b. *Perceptions of Administrative Performance: The Key to Trust in Government?* Leuven: Instituut voor de Overheid.
Van de Walle, S. 2009. 'Confidence in the Criminal Justice System: Does Experience Count?' *British Journal of Criminology*, 49, 3, 384–98.
Van de Walle, S., J.K. Kampen and G. Bouckaert. 2005. 'Deep Impact for High Impact Agencies? Assessing the Role of Bureaucratic Encounters in Evaluations of Government', *Public Performance and Management Review*, 28, 4, 532–49.
Van de Walle, S., S. Van Roosbroek and G. Bouckaert. 2008. 'Trust in the Public Sector: Is There Any Evidence for a Long-term Decline?' *International Review of Administrative Sciences*, 74, 1, 45–62.
Van Ryzin, G., D. Muzio, S. Immerwahr *et al*. 2004. 'Drivers and Consequences of Citizen Satisfaction: An Application of the American Customer Satisfaction Index to New York City', *Public Administration Review*, 64, 3, 331–41.
Van Ryzin, G.G. 2004. 'Expectations, Performance and Citizen Satisfaction with Urban Services', *Journal of Policy Analysis and Management*, 23, 3, 433–48.
Van Ryzin, G.G. 2008a. 'Citizen Perceptions of Road Smoothness: Evidence from New York with Implications for Comparative Performance Measurement', *International Review of Administrative Sciences*, 74, 4, 575–88.
Van Ryzin, G.G. 2008b. 'Validity of an On-Line Panel Approach to Citizen Surveys', *Public Performance & Management Review*, 32, 2, 236–62.
Van Ryzin, G.G. and S. Immerwahr. 2004. 'Derived Importance-performance Analysis of Citizen Survey Data', *Public Performance and Management Review*, 27, 4, 144–73.
Van Ryzin, G.G., S. Immerwahr and S. Altman. 2008. 'Measuring Street Cleanliness: A Comparison of New York City's Scorecard and Results from a Citizen Survey', *Public Administration Review*, 68, 2, 295–303.
Wansink, B. and S. Sudman. 2002. 'Predicting the Future of Consumer Panels', *Journal of Database Marketing*, 9, 4, 301.
Weiss, J.A. 1982. 'Coping with Complexity: An Experimental Study of Public Policy Decision-making', *Journal of Policy Analysis and Management*, 2, 1, 66–87.
Wentland, E.J. and K.W. Smith. 1993. *Survey Responses: An Evaluation of their Validity*. San Diego, CA: Academic Press.
Wittmer, D. 1992. 'Ethical Sensitivity and Managerial Decisionmaking: An Experiment', *Journal of Public Administration Research and Theory*, 2, 4, 443–62.
Yeric, J.L. and J.R. Tood. 1996. *Public Opinion: The Visible Politics*. Itasca, IL: F.E. Peacock.
Zaller, J. 1996. *The Nature and Origins of Mass Opinion*. Cambridge: Cambridge University Press.

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